

CNCS Support Chatbot

Setup & Installation Guide

CyberNet Computer Systems | v2.0

✓ Current Status: FULLY FUNCTIONAL

This chatbot features AI-powered responses via Ollama, a RESTful Flask API, and a SQLite database.

Overview

The CNCS Support Chatbot is a web-based customer support assistant built with Python and Flask. It allows users to browse products, look up contact information, check order status, explore FAQs, and ask free-form questions answered by a local AI model.

Component	Technology
Backend Framework	Python / Flask
Database	SQLite
AI Model	Ollama (llama3.2) — runs locally
Frontend	HTML, CSS, JavaScript
API Style	RESTful

Requirements

Before getting started, make sure you have the following installed on your machine.

1. Python 3.12+

This project was built and tested on Python 3.12.1. Earlier versions may not work correctly.

- Windows: Download from python.org. During installation, check "Add Python to PATH" — this is required.
- macOS: Python 3 can be installed via Homebrew by running: `brew install python`

To confirm Python is installed correctly, open a terminal and run:

```
python --version      (Windows)
python3 --version     (macOS)
```

PATH Warning

If you see an error like "python is not recognized", your PATH variables are not set correctly. Re-run the Python installer and ensure "Add to PATH" is checked, then restart your terminal.

2. Flask

Flask is the web framework that powers the backend. Install it using pip after Python is set up.

```
pip install Flask          (Windows)
pip3 install Flask         (macOS)
```

Need a visual walkthrough? This short video covers Flask installation in under 2 minutes:

<https://youtu.be/6EJklo3Kpr4>

3. Ollama (Local AI Model)

Ollama runs the AI language model locally on your machine. No internet connection or API key is required — everything stays on your device.

Install Ollama

- Windows / macOS: Visit ollama.com and download the installer for your operating system. Run the installer and follow the on-screen instructions.
- Linux: Open a terminal and run:

```
curl -fsSL https://ollama.com/install.sh | sh
```

Download the AI Model

Once Ollama is installed, you need to pull the llama3.2 model. Open a terminal and run:

```
ollama pull llama3.2
```

This downloads the model to your machine (~2GB). You only need to do this once.

Start the Ollama Server

Ollama must be running in the background for the chatbot's "Ask a Question" feature to work. Start it with:

```
ollama serve
```

Leave this terminal window open while using the chatbot. You can minimize it.

Note

On Windows, Ollama may start automatically after installation and run as a background service. You can check the system tray (bottom-right of your taskbar) for the Ollama icon.

4. An IDE (Recommended: VS Code)

Any code editor works. VS Code is recommended and free to download at code.visualstudio.com.

Folder Structure

Your project must be organized exactly as shown below. File names are case-sensitive on macOS and Linux.

```
your-project-folder/  
├── static/  
│   ├── chatbot.css  
│   └── chatbot.js  
├── templates/  
│   └── index.html  
├── app.py  
├── create_database.py  
├── ai-service.py  
├── config.py  
├── database.py  
├── handlers.py  
└── response_builder.py
```

Running the Application

Follow these steps in order every time you want to run the chatbot. —

Step 1 — Start Ollama

Open a terminal and start the Ollama server. Keep this window open.

```
ollama serve
```

 **Already running?**

If you see "address already in use", Ollama is already running in the background — that's fine, just move on to Step 2.

Step 2 — Set Up the Database

Open a second terminal in your project folder and run:

```
python create_database.py      (Windows)
python3 create_database.py     (macOS)
```

You should see the following confirmation message:

```
Database created and populated successfully!
```

One-time step

You only need to run `create_database.py` once, unless you want to reset the database back to its original sample data.

Step 3 — Start the Flask Server

In the same terminal, run:

```
python app.py      (Windows)
python3 app.py     (macOS)
```

You should see output like:

```
* Running on http://127.0.0.1:5000
```

Hold Ctrl and click the link in the terminal, or open your browser and navigate to:

```
localhost:5000
```

Keep this terminal open

The Flask server must stay running for the chatbot to work. Do not close this terminal window while testing.

Step 4 — Test the Chatbot

The chatbot icon appears in the bottom-right corner of the page. Click it to open and test the following features:

- Contact CNCS — Browse department contact information
- Find a Product — Browse products by category
- Check Order Status — Look up an order by number (try orders 1–5)

- Explore FAQs — Browse frequently asked questions
- Ask a Question — Type any question and get an AI-powered response (requires Ollama)

Step 5 — Shutting Down

When you are done testing:

1. Close your web browser or browser tab.
2. Click into the terminal running `app.py` and press `Ctrl + C` (Windows) or `Command + C` (macOS) to stop the Flask server.
3. If you started Ollama manually, stop it the same way in its terminal window.

If Ctrl+C does not work

Simply close the terminal window. The server will stop when the window closes.

Troubleshooting

Problem	Solution
"python is not recognized"	Python is not on your PATH. Reinstall Python and check "Add to PATH".
"No module named flask"	Flask is not installed. Run: <code>pip install Flask</code>
localhost:5000 shows nothing	Make sure <code>app.py</code> is still running in the terminal.
"Ask a Question" gives an error	Ollama is not running. Open a terminal and run: <code>ollama serve</code>
"ollama pull" is slow	The model is ~2GB. This is normal on first download — wait for it to complete.
Port 5000 already in use	Another app is using port 5000. Stop it, or change the port in <code>app.py</code> .
Database errors on startup	Delete the <code>.db</code> file and re-run <code>create_database.py</code> to reset.